MATH329 · OPERATIONS RESEARCH

Quiz 3-4 Solutions

Big-M method · 30 minutes · term 471

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01 What is Big-M and why is it needed?

Simplex starts from the slack variables because they give an identity matrix. A `≥` constraint gets a surplus with a minus sign, which cannot be a starting basic variable. The fix is to add a temporary **Artificial variable**, written `R`, to that constraint.

To make sure `R` leaves the solution and becomes zero, we penalize it in the objective with a very large coefficient `M`:

Type	Penalty in the objective
Max	`− M R`
Min	`+ M R`

The same method as the Lecture 5 slides.

02 Solution part 1: standard form

The first constraint is $\leq$, so add a slack. The second is $\geq$, so subtract a surplus.

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 \\ &\text{Subject to} \quad 4x_1 + 3x_2 - x_3 + s_1 = 24 \\ &\qquad\qquad x_1 + x_2 + 2x_3 - s_2 = 18 \\ &\qquad\qquad x_1, x_2, x_3, s_1, s_2 \geq 0 \end{aligned}$$

03 Solution part 2: Big-M

1 Add R to the second constraint and penalize it

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 - M\,R \\ &\text{Subject to} \quad 4x_1 + 3x_2 - x_3 + s_1 = 24 \\ &\qquad\qquad x_1 + x_2 + 2x_3 - s_2 + R = 18 \\ &\qquad\qquad x_1, x_2, x_3, s_1, s_2, R \geq 0 \end{aligned}$$

2 Write the tableau and fix the `Z` row

The `Z` row comes from `z − 2x₁ − 3x₂ + 5x₃ + M R = 0`:

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
s_1	4	3	−1	1	0	0	24
R	1	1	2	0	−1	1	18
Z	−2	−3	5	0	0	M	0

The problem: `R` is basic but its coefficient in the `Z` row is `M`, and it must be zero. Fix it with:

```
New Z row = Z row − M × (R row)
```

3 Run simplex as usual

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
s_1	4	3	−1	1	0	0	24
R	1	1	**2**	0	−1	1	18
Z	−M − 2	−M − 3	5 − 2M	0	M	0	−18M
s_1	$\frac{9}{2}$	$\frac{7}{2}$	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	33
x_3	$\frac{1}{2}$	$\frac{1}{2}$	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	9
Z	$-\frac{9}{2}$	$-\frac{11}{2}$	0	0	$\frac{5}{2}$	$M - \frac{5}{2}$	−45
x_2	$\frac{9}{7}$	1	0	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{1}{7}$	$\frac{66}{7}$
x_3	$-\frac{1}{7}$	0	1	$-\frac{1}{7}$	$-\frac{3}{7}$	$\frac{3}{7}$	$\frac{30}{7}$
Z	$\frac{18}{7}$	0	0	$\frac{11}{7}$	$\frac{12}{7}$	$M - \frac{12}{7}$	$\frac{48}{7}$

M is a very large number, so any coefficient containing $-M$ counts as a large negative.

Iteration 1: the most negative entry is $5 - 2M$ under x_3. The s_1 row has coefficient -1, so it is skipped, and the R row has ratio $\frac{18}{2} = 9$. So R leaves and the pivot is 2.

Iteration 2: the most negative entry is $-\frac{11}{2}$ under x_2. The ratios are $\frac{33}{\frac{7}{2}} = 9.43$ and $\frac{9}{\frac{1}{2}} = 18$. So s_1 leaves and the pivot is $\frac{7}{2}$.

All entries in the Z row are ≥ 0, and R = 0, so the solution is optimal.

Optimum $Z = \frac{48}{7} = 6.86$ at $x_1 = 0$, $x_2 = \frac{66}{7} = 9.43$, $x_3 = \frac{30}{7} = 4.29$

Check

$$Z = 3\left(\frac{66}{7}\right) - 5\left(\frac{30}{7}\right) = \frac{198}{7} - \frac{150}{7} = \frac{48}{7}$$

$$4(0) + 3\left(\frac{66}{7}\right) - \frac{30}{7} = \frac{168}{7} = 24 \qquad \leq 24 \quad \text{true}$$

$$0 + \frac{66}{7} + 2\left(\frac{30}{7}\right) = \frac{126}{7} = 18 \qquad \geq 18 \quad \text{true}$$